PII Redaction Tool - Evaluation Report

Environment Data Intern Role Assignment · Scaler AI Labs
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1. Executive Summary & Evaluation Objective

This document presents the official evaluation metrics for the hybrid PII Redaction Tool developed for Scaler AI Labs. The system combines deterministic RFC/Luhn regex detectors, contextual heuristic address models, and spaCy Named Entity Recognition (en_core_web_sm) to detect and redact sensitive PII across corporate Word documents (.docx). Metrics were generated automatically from an actual execution run against hand-annotated real prospectus data and synthetic test suites.

2. Evaluated Datasets & PII Categories

Evaluation was conducted on two distinct datasets: (1) **Real Document Dataset** (Red Herring Prospectus.docx, 32 gold annotations), and (2) **Synthetic Test Cases** (22 cases covering SSN, Credit Cards, DOB, IP addresses, and synthetic entities).

3. Overall Performance Summary

Dataset	Cases	TP	TN	FP	FN	Accuracy	Precision	Recall	F1-Score
Real Document	26	23	104	6	3	0.9338	0.7931	0.8846	0.8364
Synthetic Cases	22	21	66	6	1	0.9255	0.7778	0.9545	0.8571

4. Per-PII-Category Breakdown (Real Document)

PII Type	TP	FP	FN	Precision	Recall	F1-Score
ADDRESS	0	1	0	0.0000	1.0000	0.0000
EMAIL	8	0	0	1.0000	1.0000	1.0000
ORG	4	3	1	0.5714	0.8000	0.6666
PERSON	5	1	2	0.8333	0.7143	0.7692
PHONE	6	1	0	0.8571	1.0000	0.9231

5. Limitations & Future Enhancements

Limitations: Unanchored birth dates without keywords ('DOB:') are skipped to preserve standard financial transaction dates. Addresses spanning multi-line formatted runs without pin code anchors may be partially captured.
Future Work: Fine-tune a domain-specific Transformer model for Indian regulatory filings and integrate OCR for scanned PDFs.

6. Reproducibility

To verify this run: pytest tests/ -v and python -m src.evaluation.generate_reports.
All metrics match evaluation/results.json. Git commit: **4e5f0f75dda58ed933781c4e06f07cc00f841e91**.